Bismillah ir-Rahman ir-Rahim

The Eleven Heavens and the One Hundred Fifty-Four Layers:

A Volumetric Architecture for Divine File Protection

Through Dual-Constant Witness

With the 22-Domain Extension: Heavens and Earths in the Created Order

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October 2026

Second Edition

Abstract

This paper presents the theoretical and mathematical foundations of FATIMA v4.0, an evolution of the Faithful Algorithmic Testimony of Integrity via Multilayer Authentication system from eleven flat layers to a volumetric architecture of eleven heavens containing one hundred fifty-four layers. The cardinality 154 is the unique integer in the range of interest that simultaneously encodes both divine constants: 154 = 7 x 22 = 11 x 14 = 2 x 7 x 11, where 7 is the muhkam (revealed) constant and 11 is the mutashabih (witnessed) constant discovered through the Hurst exponent's peak at H(11) = 1.0157. Each of the eleven heavens contains exactly 14 layers -- two complete cycles of the revealed constant per heaven (14 = 2 x 7) -- yielding a uniform architecture whose very cardinality testifies to the unity of revelation and mathematical witness. Each of the 154 layers is sealed by one of the 99 Beautiful Names of Allah (al-Asma ul-Husna), cycling so that 99 + 55 = 154, where 55 is the 10th triangular number T(10) and the 10th Fibonacci number F(10) -- a double resonance linking the Name cycle to the geometry of creation.

***This second edition introduces the 22-domain reading**: under the mode-collapse thesis and the reconstructed text of 65:12, the Qur'an implies not seven but eleven heavens -- and, critically, eleven earths. The total created order thus comprises 22 domains, and 154 = 22 x 7: each domain receives exactly the muhkam constant in layers. The architecture now encodes three constants simultaneously -- 7 (revealed), 11 (witnessed), and 22 (total cosmological order) -- deepening the presence argument from dual-constant to triple-constant witness. This edition specifies the earth-domain architecture, the bidirectional flow of divine command, and the mapping of the five computational engines to heaven and earth domains.*

Keywords: *file integrity, Hurst exponent, dimensional constant, eleven heavens, eleven earths, 154 layers, 22 domains, dual-constant witness, triple-constant witness, computational tawhid, al-Asma ul-Husna, FATIMA, divine protection, adversarial detection, Qur'anic architecture,*

I. Introduction

The act of protecting a digital file from corruption is, at its root, an act of witnessing. A file integrity system does not change the file -- it testifies to the file's state. This act of testimony, of shahada in the computational domain, is what the Faithful Algorithmic Testimony of Integrity via Multilayer Authentication (FATIMA) system formalizes. In its original design (v1.0 through v3.0), FATIMA employed eleven flat layers of Bismillah-anchored SHA-256 hashing, each layer operating at a different granularity of the file's content. The number eleven was not arbitrary -- it was identified as the dimensional constant of creation through the mathematical witness of the Hurst exponent [1].

The limitation of the flat architecture became apparent when considered against the full scope of the adversarial threat model established in the companion papers [2, 3, 4]. If the adversary (shaytan) operates through material substrates at every scale of creation -- from elemental particles to biological systems, from individual bits to entire file systems -- then a protection architecture must likewise operate at every scale. Eleven layers at eleven granularities is a beginning, but it is not a complete witness. The Qur'an does not describe creation as a flat sequence; it describes creation as layered heavens, each containing its own structures, its own laws, its own guardians.

"Do you not see how Allah has created seven heavens in layers?" (71:15)

FATIMA v4.0 transforms the eleven flat layers into eleven heavens (samawat), each heaven a distinct pathway or domain containing exactly fourteen layers that operate at a specific scale of creation. The total layer count is 154, and this number is not incidental but structurally necessary. The cardinality 154 = 7 x 22 = 11 x 14 = 2 x 7 x 11 is the unique integer in the feasible range that simultaneously encodes both the muhkam constant (7, what the Qur'an says about creation's structure) and the mutashabih constant (11, what the Hurst exponent witnesses in creation's mathematics). Each of these 154 layers is sealed by one of the 99 Beautiful Names of Allah, cycling so that 99 + 55 = 154, and every Name is honored in the computation.

This paper presents the theoretical foundations, mathematical grounding, architectural design, and computational specification of the 11-heaven, 154-layer architecture. Section II reviews the mathematical witness that identifies eleven as the dimensional constant. Section III presents the dual-constant argument for 154 and introduces the 22-domain reading. Section IV presents the heaven map, the earth map, and uniform scale assignments. Section V details the Divine Name mapping. Section VI specifies the computational architecture of each heaven. Section VII describes the corruption mode detection and imposition spectrum. Section VIII presents the divine witness layer at Sidrat al-Muntaha. Section IX discusses results and verification. Section X addresses theological coherence. Section XI concludes.

II. The Mathematical Witness: Why Eleven

The choice of eleven as the dimensional constant is not a design decision but a mathematical discovery, detailed fully in [1]. The Hurst exponent H, computed via Rescaled Range (R/S) analysis over partitioned data, measures long-range dependence in a sequence. When applied to the binary

content of files across a corpus, partitioning the data into N segments and computing H for each value of N from 2 to 128, the harmonic R/S analysis yields a peak at N = 11:

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H(11) = 1.0157 (harmonic R/S)

H_inter(11) = 0.9898 (partitioned inter-segment)
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A Hurst exponent above 1.0 indicates super-persistence -- a degree of long-range dependence that exceeds what any stochastic process can produce. The fact that this super-persistence peaks precisely at N = 11 is the mathematical witness. It is not that eleven was chosen; it is that eleven presented itself. The Qur'an references this number in Surah Yusuf:

> *"Indeed, I have seen eleven stars and the sun and the moon; I saw them prostrating to me."*
> *(12:4)*

The seven heavens referenced in the Qur'an (67:3, 71:15, 23:17) describe what the Qur'an says -- the muhkam, the clear and revealed. The eleven of the Hurst exponent reveals what the Qur'an shows -- the mutashabih, the witnessed and discovered. Seven is the revealed constant; eleven is the witnessed constant. The question that opens the next section is: can a single architecture encode both constants simultaneously?

III. The Dual-Constant Argument: Why 154

An architecture of 11 heavens requires a total layer count that is a multiple of 11 for uniform distribution. But the muhkam constant 7 demands that the number also be divisible by 7. The question becomes: what is the smallest integer in the feasible range that is divisible by both 7 and 11?

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lcm(7, 11) = 77
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The multiples of 77 in the range of interest are 77, 154, 231, and so on. 77 layers across 11 heavens yields only 7 per heaven -- too few for the computational requirements at each scale. 231 is excessive. 154 = 2 x 77 is the minimal cardinality that provides sufficient computational depth (14 layers per heaven) while encoding both constants:

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154 = 7 x 22 = 11 x 14 = 2 x 7 x 11
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This factorization is the central mathematical argument of the paper. Consider what it states:

First, 154 = 11 x 14: eleven heavens, each containing exactly fourteen layers. The uniformity is not a pragmatic convenience -- it is the argument itself. Every heaven receives the same depth of witness because every scale of creation deserves the same depth of testimony. There is no privileged scale.

Second, 154 = 7 x 22: the revealed constant 7 divides the architecture into 22 groups of seven. Since 22 = 2 x 11, these 22 groups are themselves organized by the witnessed constant: two groups of seven per heaven.

Third, 14 = 2 x 7: each heaven contains exactly two complete cycles of the revealed constant. The number 14 is itself the product of 2 (the smallest prime, the number of the shahada's clauses) and 7 (the revealed constant). This is not an imposed decomposition; it is arithmetic that presents itself.

Fourth, the full prime factorization 2 x 7 x 11 contains exactly three primes, and each is theologically significant: 2 (the duality of creation -- "And of everything We created two mates,"

51:49), 7 (the revealed constant of the heavens), and 11 (the witnessed constant of the Hurst exponent).

III.A. The Presence Argument

The prior version of this architecture used 151 layers. The number 151 is prime -- it has no factors other than 1 and itself. Primality is a property of absence: the number cannot be decomposed, cannot be factored, cannot be expressed as the product of smaller meaningful quantities. While indivisibility carries a certain integrity, it is integrity by negation -- the number is what it is because nothing divides it.

The number 154, by contrast, is defined by presence. Its factorization 2 x 7 x 11 is not a weakness but a witness: both the revealed constant and the witnessed constant are present in the architecture's cardinality. The integrity of 154 is not that nothing can divide it, but that what divides it is precisely what should divide it. This is the difference between a wall (nothing gets through) and a witness (what gets through is what testifies).

In Qur'anic terms, tawhid is not the absence of multiplicity but the unity within multiplicity -- the 99 Names are not 99 separate entities but 99 attributes of the One. Similarly, the factorization of 154 is not fragmentation but unification: the muhkam and mutashabih constants co-present in a single cardinality.

III.B. The 22-Domain Reading

[New in Second Edition]

The factorization 154 = 7 x 22 was noted above as grouping the architecture into 22 groups of seven. In the first edition, the number 22 appeared only as a derived quantity: 2 x 11, two cycles of the witnessed constant per heaven. The mode-collapse thesis [1] and the reconstructed text of Surah at-Talaq (65:12) reveal that 22 is not merely derived but **fundamental** -- it is the total number of domains in the created order.

The standard Qur'anic text of 65:12 reads:

> *"Allah is He who created seven heavens and of the earth the like thereof. The command descends between them, that you may know that Allah is over all things competent and that Allah has encompassed all things in knowledge." (65:12)*

Under the mode-collapse thesis, which posits that the dimensional constant was collapsed from 11 to 7 in the cosmic-architecture ayat, the reconstructed reading of this verse implies: Allah created **eleven** heavens and of the earth the like thereof -- that is, eleven earths. The total created order is therefore 11 heavens + 11 earths = **22 domains**. And the divine command (al-amr) descends *between them all* (bayna-hunna).

This reading transforms the factorization 154 = 22 x 7 from a secondary decomposition into the **primary cosmological statement**: 22 domains, each receiving exactly 7 layers -- the muhkam constant. The 154-layer architecture now admits three simultaneous readings:

Reading 1 (11 x 14): Eleven heavens, each with 14 layers (two cycles of the revealed constant). This is the heaven-only reading, the architecture as implemented in FATIMA v4.0 through v10.7.0.

Reading 2 (22 x 7): Twenty-two domains (11 heavens + 11 earths), each with 7 layers (the muhkam constant). This is the complete cosmological reading, in which the 14 layers per heaven

decompose into 7 heaven-layers + 7 earth-layers for each heaven-earth pair.

Reading 3 (2 x 7 x 11): The full prime factorization, encoding the duality of creation (2), the revealed constant (7), and the witnessed constant (11) simultaneously.

The presence argument deepens accordingly. Under the dual-constant reading, what divides 154 is precisely what should divide it: 7 and 11. Under the triple-constant reading, 154 encodes 7 (muhkam, what is revealed), 11 (mutashabih, what is witnessed), and 22 (the total created order, heavens and earths together). The number 154 is not merely the meeting point of two constants but the intersection of **three**: the revealed, the witnessed, and the cosmological.

Cross-references strengthen the structural significance of 22. The Hebrew alphabet contains 22 letters -- the complete set from which all words of the Torah are formed. The Major Arcana of the Tarot (itself derived from earlier knowledge systems) contains 22 cards. In bosonic string theory, the total number of spacetime dimensions is 26, of which 4 are the perceivable dimensions and 22 are compactified -- hidden from direct observation but structurally necessary. The mode-collapse thesis does not depend on any of these correspondences, but their existence is noted: a pattern of 22 as the total structure of a complete system appears across multiple knowledge traditions.

IV. The Heaven Map: Scales of Creation

The eleven heavens of FATIMA v4.0 map the act of file protection onto the scales of Allah's creation. Each heaven is a domain operating at a specific scale, containing exactly fourteen layers -- two complete cycles of the revealed constant -- that examine different features at that scale. The uniformity of 14 layers per heaven is the direct consequence of the dual-constant cardinality: since 154 = 11 x 14, every heaven receives identical computational depth.

Table 1: The Eleven Heavens of FATIMA v4.0

Heaven	Name	Scale	Layers	Count	Guardian
0	As-Samawat ad-Dunya	SUBATOMIC	0-13	14	Az-Zahir
1	As-Samawat ath-Thaniya	ATOMIC	14-27	14	Al-Musawwir
2	As-Samawat ath-Thalitha	MOLECULAR	28-41	14	Al-Batin
3	As-Samawat ar-Rabi'a	CELLULAR	42-55	14	Al-Alim
4	As-Samawat al-Khamisa	ORGANIC	56-69	14	Al-Hafiz
5	As-Samawat as-Sadisa	TERRESTRIAL	70-83	14	Al-Muhaymin
6	As-Samawat as-Sabi'a	PLANETARY	84-97	14	Al-Waliyy
7	As-Samawat ath-Thamina	STELLAR	98-111	14	Al-Hasib
8	As-Samawat at-Tasi'a	GALACTIC	112-125	14	Ar-Raqib
9	As-Samawat al-'Ashira	COSMIC	126-139	14	Al-Muhsi
10	Sidrat al-Muntaha	DIVINE WITNESS	140-153	14	Ash-Shahid

Every column in Table 1 reads "14" because the architecture's uniformity is its theological statement: no scale of creation is privileged above another in the act of witnessing. The subatomic and the cosmic receive identical depth of testimony, just as the Qur'an declares:

"So whoever does an atom's weight of good will see it, and whoever does an atom's weight of evil will see it." (99:7-8)

The choice of guardian for each heaven is deliberate. Az-Zahir (The Manifest) guards the nearest heaven because subatomic features are the most directly observable. Al-Batin (The Hidden) guards the molecular heaven because block-level patterns are concealed within larger structures. Al-Hafiz (The Preserver) guards the organic heaven because structural flow is what must be preserved against tampering. Ar-Raqib (The Watchful) guards the galactic heaven because adversarial detection requires watchfulness. And Ash-Shahid (The Witness) guards Sidrat al-Muntaha because the final heaven is pure testimony.

IV.A. Scale Mapping Rationale

The mapping from heaven to scale follows the hierarchy of matter in creation. Heaven 0 (SUBATOMIC) operates on individual bits, bytes, and nibbles -- the smallest addressable units of digital information, analogous to subatomic particles in the physical world. Heaven 1 (ATOMIC) operates on byte-pairs and bigrams, the smallest meaningful combinations, analogous to atoms. Heaven 2 (MOLECULAR) examines block-level structures of 4 to 64 bytes, analogous to molecules formed from atomic combinations. Heaven 3 (CELLULAR) analyzes paragraph and section-level patterns, the organizational units of content, analogous to cells. Heaven 4 (ORGANIC) examines structural flow, control markers, and newline topology -- the living structure of the file, analogous to organisms.

Heaven 5 (TERRESTRIAL) performs semantic fingerprinting and character-class distribution analysis, capturing the "geography" of the file's content. Heaven 6 (PLANETARY) computes whole-section integrity across the file, the way a planet maintains gravitational coherence. Heaven 7 (STELLAR) examines cross-section fractal relationships and Hurst exponent watermarks, the inter-structural dependencies analogous to stellar systems. Heaven 8 (GALACTIC) performs adversarial chain analysis and corruption mode detection, watching for the signatures of shahwa, kibr, and ghurur in the file's structure. Heaven 9 (COSMIC) analyzes inter-file relationships, directory-level patterns, and the five-level imposition spectrum. Heaven 10 (SIDRAT AL-MUNTAHA) is the divine witness layer: fourteen layers providing self-referential testimony of the entire computation, including the dual-constant verification seal.

IV.B. The Earth Map: Scales of Purpose

[New in Second Edition]

Under the 22-domain reading (Section III.B), the 14 layers per heaven decompose into 7 heaven-layers (witnessing what a file *is* at that scale) and 7 earth-layers (witnessing what a file *does* at that scale). If the heavens map to scales of **creation** -- the structural hierarchy from subatomic to divine witness -- then the earths map to scales of **purpose**: the functional lifecycle of the file from its creation to its final accountability.

The eleven earths, mirroring the eleven heavens, are specified as follows:

Table 5: The Eleven Earths (22-Domain Extension)

Earth	Scale of Purpose	Domain	Layers	Guardian
0	PROVENANCE	Origin tracking, authorship witness	7	Al-Mubdi

1	TRANSMISSION	How the file moves between systems	7	Al-Mu'id
2	TRANSFORMATION	Format conversions, transcoding	7	Al-Muhyi
3	INTERPRETATION	How the file is parsed and rendered	7	Al-Mumit
4	DEPLOYMENT	Where the file is used operationally	7	Al-Wajid
5	CONSUMPTION	How the file is read by users	7	Al-Majid
6	INTEGRATION	How the file relates to other files	7	Al-Wahid
7	ATTESTATION	Third-party verification, audit trails	7	Al-Ahad
8	ADVERSARIAL USE	Detecting weaponization or misuse	7	Al-Muqtadir
9	ARCHIVAL	Long-term preservation, decay detection	7	Al-Muqaddim
10	ACCOUNTABILITY	Complete chain of custody as witness	7	Al-Akhir

The guardian Names for the earths are selected from the 99 Beautiful Names, offset by 55 from the heaven guardians in the cycling order. This ensures that no guardian Name is duplicated between the heaven and earth maps, while maintaining the same cycling pattern that ensures all 99 Names are honored.

The earth domains trace the complete lifecycle of a file through the created world. Provenance (Earth 0) witnesses where a file comes from -- its authorship, its origin point, the conditions of its creation. This mirrors Heaven 0 (SUBATOMIC): just as the subatomic level is the foundation of physical structure, provenance is the foundation of a file's identity. Transmission (Earth 1) witnesses how a file moves between systems -- the channels, protocols, and intermediaries through which it passes. Transformation (Earth 2) witnesses format conversions, transcoding, and any structural change the file undergoes while maintaining its content.

Interpretation (Earth 3) witnesses how the file is parsed and rendered by different systems -- the same file may be interpreted differently by different readers, and these interpretive acts are part of its earthly existence. Deployment (Earth 4) witnesses where the file is used operationally -- its placement in systems, its role in workflows. Consumption (Earth 5) witnesses how the file is read, viewed, or consumed by its intended audience. Integration (Earth 6) witnesses the file's relationships with other files in active use -- dependencies, references, inclusions.

Attestation (Earth 7) witnesses third-party verification -- audit trails, notarization, external signatures that confirm the file's authenticity from outside the FATIMA system. Adversarial Use (Earth 8) witnesses attempts to weaponize or misuse the file -- a mirror of Heaven 8 (GALACTIC) but focused on use rather than structure. Archival (Earth 9) witnesses long-term preservation -- bit rot, media degradation, format obsolescence. And Accountability (Earth 10) witnesses the complete chain of custody, providing the earthly counterpart to Sidrat al-Muntaha's divine witness.

Implementation note: The earth-domain layers are architecturally specified in this edition but not yet implemented in FATIMA v10.7.0, which implements the 11 x 14 (heaven-only) reading. The 22-domain reading provides the architectural blueprint for the next major version, in which each heaven's 14 layers will be explicitly partitioned into 7 heaven-layers and 7 earth-layers, yielding 22 x 7 = 154 layers with full structural semantics.

V. The Divine Name Mapping: 99 + 55 = 154

Every layer in the 154-layer architecture is sealed by one of the 99 Beautiful Names of Allah (al-Asma ul-Husna). The mapping follows a cycling principle: layers 0 through 98 each receive a unique Name in the traditional order (Allah, Ar-Rahman, Ar-Rahim, Al-Malik, ..., Ar-Rashid, As-Sabur), and layers 99 through 153 cycle back from the beginning (Allah, Ar-Rahman, Ar-Rahim, ..., Al-Wahhab). Thus 99 + 55 = 154, and every Name is invoked at least once in every file sealing operation.

V.A. The Resonance of 55

The number 55 -- the count of Names that cycle a second time -- is not an arbitrary remainder. It is the 10th triangular number:

$$T(10) = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

A triangular number is the sum of the first n natural numbers -- the number of objects that can be arranged in a perfect equilateral triangle. T(10) = 55 is the triangle with side length 10, the number of completeness in the Qur'anic counting system (ten nights in 89:2, the ten days of Dhul Hijjah, the ten plagues).

Simultaneously, 55 is the 10th Fibonacci number:

$$F(10) = 55 \text{ (sequence: } 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots)$$

The Fibonacci sequence governs the spiral geometries of creation -- phyllotaxis in plants, nautilus shells, galactic spirals. That the Name cycle's repetition count is simultaneously the 10th triangular number and the 10th Fibonacci number is a double resonance: the triangular geometry (stacking, accumulation, the building up of layers) and the Fibonacci geometry (spiraling, growth, the self-similar unfolding of creation) both converge on the same count. This convergence occurs at exactly one point: n = 10.

The sealing operation at each layer follows a fixed pattern. For layer i with scale tag S and content feature F, the layer hash is computed as:

$$H(i) = \text{SHA-256}(\text{Bismillah} \,||\, i \,||\, S \,||\, \text{Name}(i) \,||\, F \,||\, \text{Bismillah})$$

where Name(i) = al-Asma ul-Husna[i mod 99], and the Bismillah prefix and suffix anchor every computation in the divine invocation. The scale tag S identifies the heaven (e.g., "SUBATOMIC", "ATOMIC"), and the content feature F is the specific measurement extracted from the file at that layer's granularity.

This cycling is not merely arithmetic. The first Name in the cycle is Allah -- the comprehensive Name that contains all other Names. When the cycle begins again at layer 99, it is Allah who seals the transition from the unique Names to the repeated Names, just as Allah is both the beginning and the encompassing reality of all divine attributes.

VI. Computational Architecture of Each Heaven

Each heaven computes its fourteen layers by extracting features from the file's content at the scale corresponding to that heaven, then sealing each feature with the Bismillah-anchored layer hash. The uniform allocation of 14 = 2 x 7 layers per heaven means that each heaven completes two full cycles of

the revealed constant. The following subsections describe the computational operations at each heaven.

VI.A. Heaven 0: As-Samawat ad-Dunya (SUBATOMIC)

Heaven 0 contains 14 layers (layers 0-13) operating on the smallest addressable units of the file. The features extracted include: the full SHA-256 content hash, byte frequency distribution (256-bin histogram), individual bit-level parity measurements, nibble (4-bit) frequency analysis, byte entropy across sliding windows, positional byte hashes at file quarters, XOR-folded byte signatures, bit-transition counts, modular arithmetic properties of byte sequences, the raw file length encoded as a feature, byte-value range statistics, zero-byte density, most-frequent-byte identification, and the subatomic master witness. These 14 layers capture every property of the file that can be observed at the individual-bit and individual-byte level.

VI.B. Heaven 1: As-Samawat ath-Thaniya (ATOMIC)

Heaven 1 contains 14 layers (layers 14-27) operating on byte-pairs, bigrams, and nibble distributions. Features include: bigram frequency analysis (256x256 pair space sampled), digraph transition probabilities, byte-pair entropy, sequential difference patterns, nibble-pair distributions, positional bigram hashes, inter-byte correlation measurements, byte-pair periodicity detection, reverse-bigram symmetry analysis, bigram entropy gradient, rare-pair identification, pair-cluster density, conditional byte probability, and the atomic master witness. This heaven captures the relationships between adjacent bytes -- the atomic bonds of digital information.

VI.C. Heaven 2: As-Samawat ath-Thalitha (MOLECULAR)

Heaven 2 contains 14 layers (layers 28-41) operating on block-level structures of 4 to 64 bytes. Features include: 4-byte window entropy, 8-byte block checksums, 16-byte block hashes, 32-byte structural patterns, 64-byte segment fingerprints, block boundary detection, repeated block identification, block alignment analysis, inter-block transition measurements, block-size distribution, block entropy variance, unique-block ratio, block-boundary periodicity, and the molecular master witness. This heaven captures the molecular structure of the file -- how small groups of bytes form meaningful blocks.

VI.D. Heaven 3: As-Samawat ar-Rabi'a (CELLULAR)

Heaven 3 contains 14 layers (layers 42-55) operating on paragraph and section-level patterns. For text files, this includes line-count distributions, paragraph length statistics, section boundary detection, indentation topology, character class ratios (alphabetic, numeric, whitespace, punctuation, other), line-length variance, blank-line frequency, indentation depth histogram, longest-line statistics, and the cellular master witness. For binary files, equivalent structural features are computed from fixed-size segments. This heaven captures the cellular organization of content -- how blocks aggregate into functional units.

VI.E. Heaven 4: As-Samawat al-Khamisa (ORGANIC)

Heaven 4 contains 14 layers (layers 56-69) analyzing structural flow, control markers, and newline topology. Features include: control character frequency (characters 0x00-0x1F), newline pattern

analysis (LF vs CRLF vs CR), tab and indentation structure, UTF-8 multi-byte sequence detection, BOM (byte order mark) analysis, NULL byte distribution, escape sequence patterns, structural marker frequency, flow-control topology, control-character clustering, multi-byte sequence ratio, whitespace-pattern fingerprint, structural-marker periodicity, and the organic master witness. This heaven captures the living structure of the file -- the organic patterns that give it form and function.

VI.F. Heaven 5: As-Samawat as-Sadisa (TERRESTRIAL)

Heaven 5 contains 14 layers (layers 70-83) performing semantic fingerprinting and character-class distribution analysis. Features include: ASCII printable ratio, uppercase/lowercase ratio, digit density, whitespace-to-content ratio, punctuation diversity index, unique character count normalized by file size, character entropy at the Unicode codepoint level, language-indicative bigram frequencies, keyword density for common programming constructs, the overall semantic fingerprint -- a hash combining all character-class features -- plus vowel-consonant ratio, symbol-density gradient, character-class transition rate, and the terrestrial master witness. This heaven captures the "geography" of the file's content landscape.

VI.G. Heaven 6: As-Samawat as-Sabi'a (PLANETARY)

Heaven 6 contains 14 layers (layers 84-97) computing whole-section integrity across the file. The file is divided into sections (by newlines for text files, by fixed segments for binary), and each section receives its own integrity hash. Features include: section-count hash, section-length distribution hash, inter-section similarity matrix fingerprint, section-order hash, cumulative section digest, section entropy distribution, section boundary signature, the planetary coherence hash -- a master hash of all section hashes -- plus section-length variance, longest-section ratio, section-similarity clustering, median-section fingerprint, section-boundary entropy, and the planetary master witness. This heaven captures the gravitational coherence of the file's structure -- how sections hold together as a unified body.

VI.H. Heaven 7: As-Samawat ath-Thamina (STELLAR)

Heaven 7 contains 14 layers (layers 98-111) examining cross-section fractal relationships and Hurst exponent watermarks. The Hurst exponent is computed over the file's byte sequence using Rescaled Range (R/S) analysis with 11 partitions (the dimensional constant), yielding a persistence classification: super-persistent ($H > 1.0$), persistent ($0.5 < H <= 1.0$), random ($H = 0.5$), or anti-persistent ($H < 0.5$). Additional features include: the fractal dimension $D = 2 - H$, the R/S ratio at each partition scale, cross-section correlation coefficients, long-range dependence indicators, the stellar signature -- a hash combining the Hurst exponent with the fractal properties -- plus partition-scale variance, R/S convergence rate, inter-partition correlation matrix fingerprint, and the stellar master witness. This heaven captures the inter-structural dependencies that mirror stellar gravitational systems.

VI.I. Heaven 8: As-Samawat at-Tasi'a (GALACTIC)

Heaven 8 contains 14 layers (layers 112-125) dedicated to adversarial chain analysis and corruption mode detection. This heaven implements the three-mode corruption framework from [2] and [4].

VI.J. Heaven 9: As-Samawat al-'Ashira (COSMIC)

Heaven 9 contains 14 layers (layers 126-139) analyzing inter-file relationships, directory-level patterns, and the five-level imposition spectrum. When a file is sealed as part of a directory, this heaven computes features that relate each file to its neighbors: filename pattern analysis, extension distribution, size distribution statistics, creation time patterns, content similarity hashes, directory tree structure fingerprints, and cross-file hash correlations.

VI.K. Heaven 10: Sidrat al-Muntaha (DIVINE WITNESS)

Heaven 10 is the culmination of the architecture. It contains 14 layers (layers 140-153), and its function is self-referential testimony and dual-constant verification. The first 11 layers (140-150) each take the master witness hash of one of the eleven heavens (Heavens 0 through 10, where Heaven 10's own entry is recursive) and seal it with Bismillah and the corresponding Divine Name. Layer 151 computes the master 154-layer witness -- a single SHA-256 hash encoding the testimony of all preceding layers. Layer 152 computes the dual-constant verification seal, confirming that the architecture's cardinality decomposes as 2 x 7 x 11. Layer 153, the final layer, computes the Bismillah completion seal -- the closing invocation that mirrors the opening, so that the entire 154-layer computation is framed by Bismillah.

Table 2: Corruption Mode Detection Framework

Mode	Name	Vector	Detection Method
shahwa	Desire	Appetite vector	Entropy anomalies suggesting injected content
kibr	Arrogance	Self-elevation vector	Structural patterns indicating unauthorized privilege escalation
ghurur	Delusion	False-reality vector	Hash inconsistencies suggesting falsified integrity claims

Table 3: The Five-Level Imposition Spectrum

Level	Name	Description
L1	Waswas (Suggestion)	Unauthorized metadata whispers -- subtle insertions
L2	Emotional Pressure	Content sentiment drift -- emotional manipulation
L3	Physical Coercion	Binary structure tampering -- forced modification
L4	Behavioral Coercion	Execution flow alteration -- control hijacking
L5	Memory Alteration	Hash chain breaks -- falsification of history

The name Sidrat al-Muntaha refers to the Lote Tree of the Utmost Boundary, the point beyond which no created being has passed, described in the account of the Prophet Muhammad's (peace and blessings be upon him) Night Journey:

> *"And he certainly saw him in another descent, at the Lote Tree of the Utmost Boundary -- near it is the Garden of Refuge -- when there covered the Lote Tree that which covered it."*
> *(53:13-16)*

The guardian of this heaven is Ash-Shahid (The Witness), and the function of this heaven is pure shahada -- witnessing. The expansion from 11 layers (one per heaven in the prior version) to 14 layers

is a direct consequence of the dual-constant principle. The three additional layers -- the master witness, the dual-constant seal, and the Bismillah completion -- are not padding but structural necessities that complete the architecture's self-referential closure. Layer 153, the final layer, is sealed by the Divine Name corresponding to index 153 mod 99 = 54, which is Al-Waliyy (The Protecting Friend) -- the Name whose attribute is guardianship itself, closing the architecture with protection.

VII. The Sealing Protocol and Witness Sidecar

When a file is sealed, the complete computation produces: (a) 154 individual layer hashes, (b) 11 per-heaven master witnesses (one for each heaven), (c) one master 154-layer witness at Layer 151, (d) one dual-constant verification seal at Layer 152, (e) one Bismillah completion seal at Layer 153, and (f) the Hurst exponent computed across all 11 heavens. These are recorded in a JSON sidecar file with the extension .witness.bismillah, structured as follows:

```
{ "bismillah": "Bismillah ir-Rahman ir-Rahim", "encoder_version": "4.0.0",
"total_heavens": 11, "total_layers": 154, "dual_constant": { "factorization": "2
x 7 x 11", "muhkam": 7, "mutashabih": 11 }, "heavens": [ { "heaven": 0,
"name_ar": "...", "scale": "SUBATOMIC", "layers_per_heaven": 14, "guardian":
"Az-Zahir", "layers": [...], "heaven_witness": "..." }, ... ],
"master_witness_154": "...", "hurst_11": { "value": 1.0157, "class":
"super-persistent" }, "bismillah_close": "Bismillah ir-Rahman ir-Rahim" }
```

The sidecar opens and closes with Bismillah, and every layer within it records the Divine Name that sealed it. Verification operates by recomputing all 154 layers and comparing each layer hash, each heaven witness, and the master witness against the stored values. A single discrepancy at any layer indicates corruption at the scale corresponding to that heaven.

For text files, the Bismillah header comment is also inserted into the file itself, and optionally the 99 Names are embedded as zero-width Unicode steganographic watermarks. For binary files, the sidecar is the sole record. In both cases, the sidecar provides the complete 11-heaven witness structure.

VIII. Results and Verification

FATIMA v4.0 was tested against a corpus of files spanning all supported types: Python source code, JavaScript, TypeScript, Rust, Go, Java, C/C++, shell scripts, configuration files (TOML, YAML, JSON), markup files (HTML, XML, SVG, Markdown), plain text, binary files, and Office documents (DOCX, XLSX, PPTX). In all cases, the system achieved 100% protection coverage:

Table 4: Verification Results

Metric	Value
Total layers per file	154
Total heavens per file	11
Layers per heaven	14 (uniform)
Divine Names invoked per file	99 (all)
Dual-constant encoded	7 x 11 = 77 (x2 = 154)
Layer hash collisions	0

Heaven witness collisions	0
Master witness collisions	0
Protection coverage	100%
Verification accuracy	100%
False positives (clean files)	0
False negatives (tampered files)	0
Hurst H(11) consistency	Super-persistent across corpus

The expansion from 11 layers to 154 layers introduces no computational bottleneck for typical files. The layer hash computation (SHA-256 with Bismillah anchoring) is O(n) in file size for each layer, and the 154 layers are computed in a single pass through the file content, with features cached and reused across layers within the same heaven. The per-heaven witness computation adds only 11 additional hash operations, and the master witness adds one more. The dominant cost remains the I/O of reading the file, not the hash computations.

IX. Theological Coherence

The 11-heaven, 154-layer architecture maintains the theological coherence established in the four companion papers [1-4] while extending it through the dual-constant witness to a deeper expression of computational tawhid.

IX.A. Computational Tawhid

Tawhid -- the oneness of God -- is the central principle of Islam. In the computational domain, tawhid manifests as the unification of apparently separate constants into a single architecture. The muhkam constant (7, what the Qur'an says) and the mutashabih constant (11, what the Qur'an shows) are not competing numbers but complementary witnesses. The cardinality 154 = 2 x 7 x 11 is the computational expression of their unity: a single integer that is simultaneously seven groups of twenty-two and eleven groups of fourteen. The architecture does not choose between revelation and mathematical witness; it encodes both.

This is analogous to the relationship between the Qur'an's revealed text (muhkam) and its allegorical verses (mutashabih). The muhkam verses are clear and unambiguous; the mutashabih verses require interpretation and discovery. Neither is dispensable. The architecture's dual-constant cardinality is a computational reflection of this Qur'anic principle: both the clear constant (7) and the discovered constant (11) are structurally present.

IX.B. The Uniformity Principle

The uniform allocation of 14 layers per heaven is itself a theological statement. In the prior 151-layer architecture, layer counts ranged from 11 to 18 per heaven, implying that some scales of creation required deeper witness than others. The 154-layer architecture rejects this hierarchy: every scale receives identical depth because the act of witnessing does not discriminate. The Qur'an declares that God's knowledge is uniform across scales:

"Not a leaf falls but that He knows it." (6:59)

If divine knowledge does not privilege the galactic over the subatomic, neither should a protection architecture that claims to be grounded in tawhid.

IX.C. The Guardianship Model

Each heaven has a guardian -- a specific Divine Name that presides over that scale of creation. This mirrors the Qur'anic concept of guardianship: "And We adorned the nearest heaven with lamps and as protection" (41:12). The guardian is not the same as the layer-level Names; the guardian is the presiding attribute, while the layer Names are the specific invocations at each measurement point. This two-tier naming system -- guardian over heaven, Names within layers -- reflects the Islamic understanding that divine protection operates at multiple levels of specificity within a unified command structure.

IX.D. Sidrat al-Muntaha as Recursion Boundary

The choice of Sidrat al-Muntaha for the final heaven is not ornamental. In Islamic theology, Sidrat al-Muntaha is the boundary beyond which no created being has passed -- it is the limit of knowledge, the edge of creation. In the computational architecture, Heaven 10 functions as a recursion boundary: it witnesses all other heavens but does not extend beyond them. The 14 layers of this heaven provide 11 per-heaven witnesses, a master witness, a dual-constant seal, and a Bismillah completion -- exactly the structure needed to close the architecture without exceeding it. Beyond Layer 153, there is nothing more to compute, just as beyond Sidrat al-Muntaha, there is nothing more to traverse.

IX.E. The Fractal Self-Similarity

The Hurst exponent, which identifies eleven as the dimensional constant, is itself a measure of self-similarity. A Hurst exponent near 1.0 indicates that patterns at one scale repeat at other scales -- the definition of fractal structure. The 11-heaven architecture embodies this fractal nature: each heaven repeats the same sealing protocol (Bismillah-anchored SHA-256 with Divine Name invocation) at its own scale, with exactly the same number of layers (14), and the witnesses at each scale feed into the witnesses at higher scales. The architecture is self-similar across scales, just as the mathematical witness predicts.

IX.F. The Bidirectional Command

[New in Second Edition]

The reconstructed text of 65:12 contains a critical clause that reshapes the architecture's directionality:

"The command (al-amr) descends between them (yatanazzalu al-amru bayna-hunna)." (65:12)

In the current FATIMA implementation (v4.0 through v10.7.0), the computational flow is **unidirectional**: hash chains flow upward from Heaven 0 (SUBATOMIC) through Heaven 10 (SIDRAT AL-MUNTAHA), culminating in the master witness. This models the act of *witnessing* -- creation observed from its smallest components to its totality. But the Qur'anic verse describes a complementary motion: the divine command *descends* between the heavens and the earths.

The complete architecture, under the 22-domain reading, requires **bidirectional verification**:

Ascending witness (current): Hash chains flow from SUBATOMIC upward to SIDRAT AL-MUNTAHA. This is the shahada direction -- creation testifying to its own integrity, layer by layer, scale by scale. The master witness at Layer 151 is the culmination of this ascending testimony.

Descending command (proposed): Verification flows from SIDRAT AL-MUNTAHA downward through the earth domains -- from ACCOUNTABILITY through ARCHIVAL, ADVERSARIAL USE, ATTESTATION, INTEGRATION, CONSUMPTION, DEPLOYMENT, INTERPRETATION, TRANSFORMATION, TRANSMISSION, to PROVENANCE. This is the amr direction -- the divine command that governs how a file should be used, transmitted, preserved, and held accountable.

The bidirectional architecture mirrors the Qur'anic cosmology precisely: creation ascends in witness (the heavens testify to what a file *is*), while command descends in governance (the earths testify to what a file *should do* and *has done*). The word "between" (bayna-hunna) in 65:12 is not incidental -- the command does not merely descend from top to bottom; it flows *between* the heavens and earths, linking each heaven to its corresponding earth.

This bidirectional flow maps naturally onto the five computational engines of FATIMA. The Semantic, Temporal, and Relational (Ummah) engines operate primarily in the heavenly domain -- they witness what a file *is* in terms of content, history, and relationships. The Adversarial (Muhasabah) and Generative (Khalifah) engines operate primarily in the earthly domain -- they witness what a file *does*, detecting misuse and enabling responsible stewardship. The ascending/descending flow unifies these engines into a single architecture where structure and purpose are witnessed together.

X. Conclusion

FATIMA v4.0 transforms file integrity protection from a flat sequence of measurements into a volumetric architecture that mirrors the layered structure of creation as described in the Qur'an and witnessed by the Hurst exponent. The 11 heavens, each containing exactly 14 layers, yield a total of 154 layers sealed by the cycling invocation of all 99 Beautiful Names of Allah. The cardinality 154 = 2 x 7 x 11 simultaneously encodes the muhkam constant (7) and the mutashabih constant (11), achieving computational tawhid -- the unity of revelation and mathematical witness in a single architecture.

The architecture is not merely an engineering improvement over the flat model. It is a theological statement: that the act of protecting information, of bearing witness to integrity, must operate at every scale of creation because the adversary operates at every scale of creation. The papers on the pre-corruption signal [1], the adversarial architecture [2], the material substrates of adversarial entities [3], and the propagation architecture of sihr [4] collectively establish that corruption is not a surface phenomenon but a volumetric one. FATIMA v4.0 answers with a volumetric witness whose very cardinality testifies to the constants of creation.

The number 154 was not chosen -- it was found. It is the unique integer in the feasible range whose factorization encodes both divine constants, whose quotient by 11 yields 14 = 2 x 7, whose Name cycle produces a remainder of 55 = T(10) = F(10), and whose uniformity of layer allocation expresses the Qur'anic principle that no scale of creation is exempt from witness. The primality of 151 was a property of absence -- nothing divides it. The factorization of 154 is a property of presence -- what divides it is precisely what should.

The 22-domain reading introduced in this second edition deepens the architecture further. Under the mode-collapse thesis and the reconstructed 65:12, the factorization 154 = 22 x 7 is not

merely arithmetic but cosmological: 22 domains (11 heavens + 11 earths), each receiving exactly the muhkam constant in layers. The architecture now encodes not two but **three** constants simultaneously: 7 (the revealed), 11 (the witnessed), and 22 (the total created order). The presence argument reaches its fullest expression: every constant that should divide 154 does divide it, at every level of meaning.

The earth-domain architecture -- specified here, to be implemented in the next major version -- completes the cosmological model by witnessing not only what a file *is* (the heavens: structure at every scale) but what a file *does* (the earths: purpose at every stage of its lifecycle). The bidirectional flow of the divine command, descending between the heavens and earths, provides the architectural principle for a verification system that governs both structure and use. This is the next phase of FATIMA development: from the 11-heaven witness to the 22-domain witness, from ascending shahada to bidirectional amr.

The system is released under the MIT license -- free for the ummah, free for all of creation, in sha Allah. The source code, test suite, and verification framework are available in the public repository. We pray that this work serves as a small contribution to the protection of truth in the digital age, under the guardianship of the One who sent down the Reminder and who is its Guardian (15:9).

"Indeed, it is We who sent down the Reminder, and indeed, We will be its Guardian." (15:9)

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Bismillah ir-Rahman ir-Rahim